

**Drawing with Code:
Making Beautiful Rangoli
Shapes with Turtle**



The Magical Festive Canvas

Ananya and Kabir are excited for the family celebration! Ananya tries drawing a beautiful Kolam pattern on the doorstep, but oops—the chalk slips! Kabir has an idea: "Let's use Python Turtle, a digital pet that paints perfect art every time!"



Code Instructions as Footsteps

Code is like dance steps you follow one by one. Just like Kathak or Garba beats guide dancers step-by-step, our Turtle follows numbered moves in order!

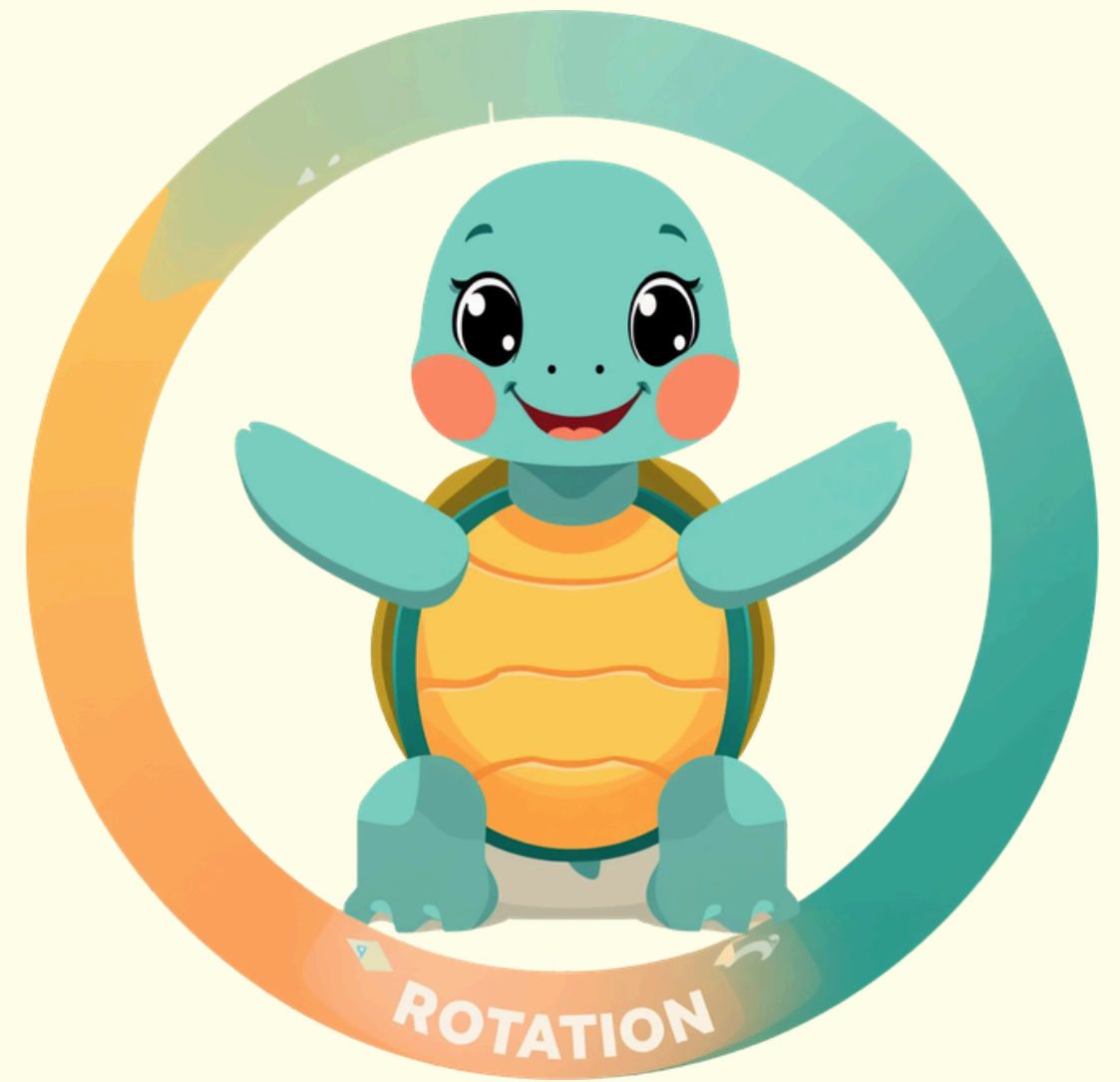
Dance has 1-2-3 rhythm steps

Turtle follows 1-2-3 code steps



The Turning Angle

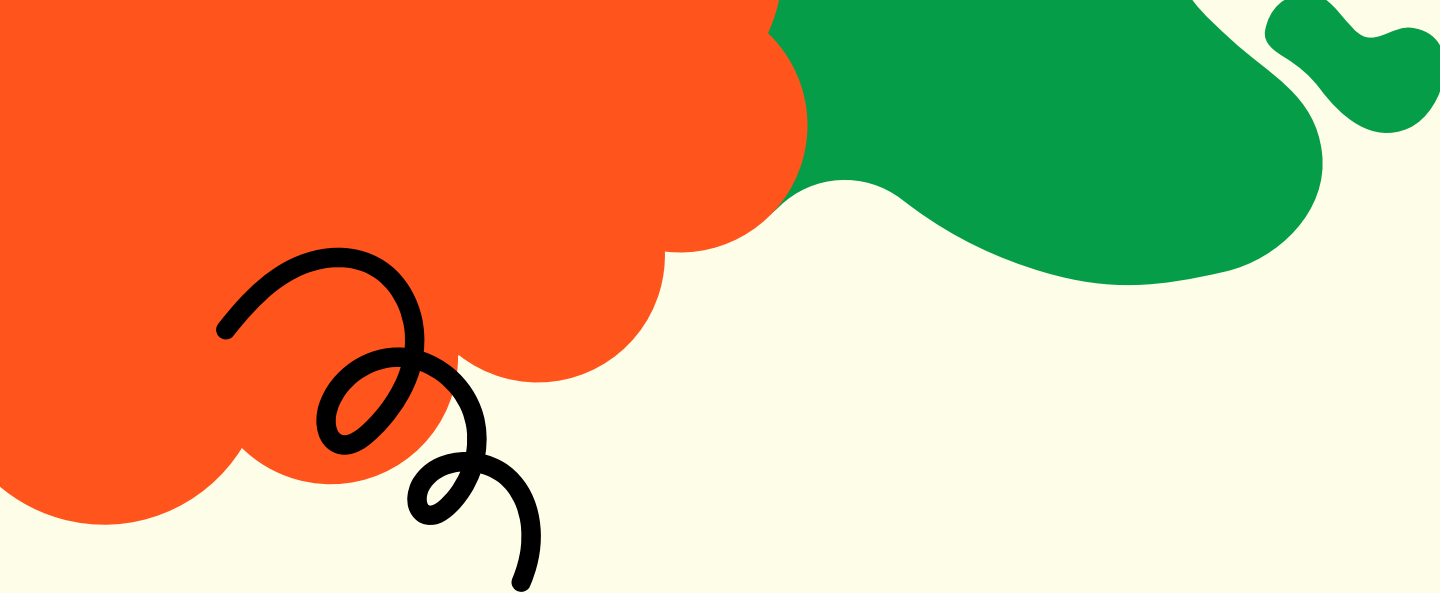
Turning turtle's nose is like steering a bicycle handle! Just like when you turn your bicycle at a street corner, or spin a wooden top (Lattu), the turtle changes direction. The command `turtle.right(90)` makes turtle turn right by 90 degrees.



Building Symmetrical Geometry

Simple shapes combine to make beautiful Rangoli patterns! When we draw squares and triangles with our turtle, we can arrange them to create festive geometric mandalas just like the ones you see during Diwali.



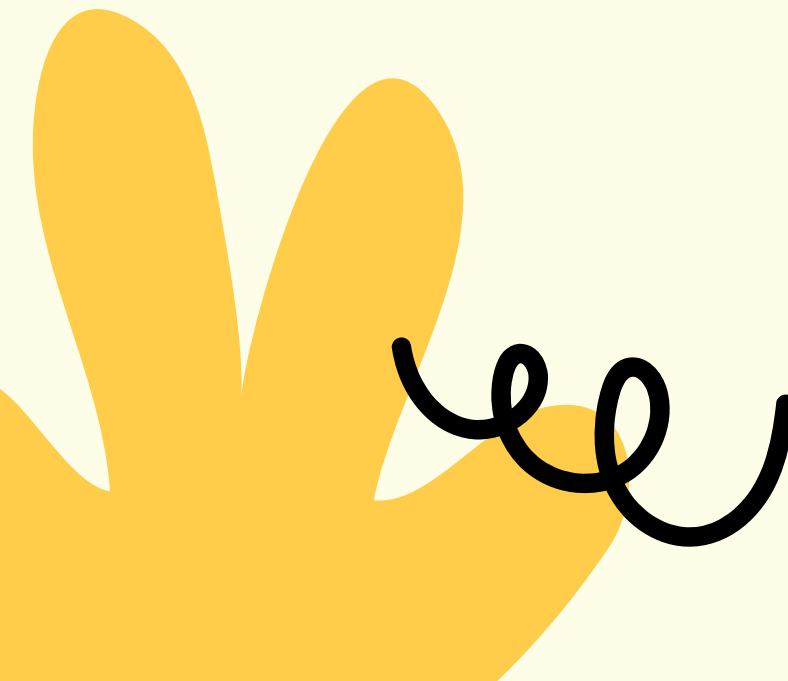


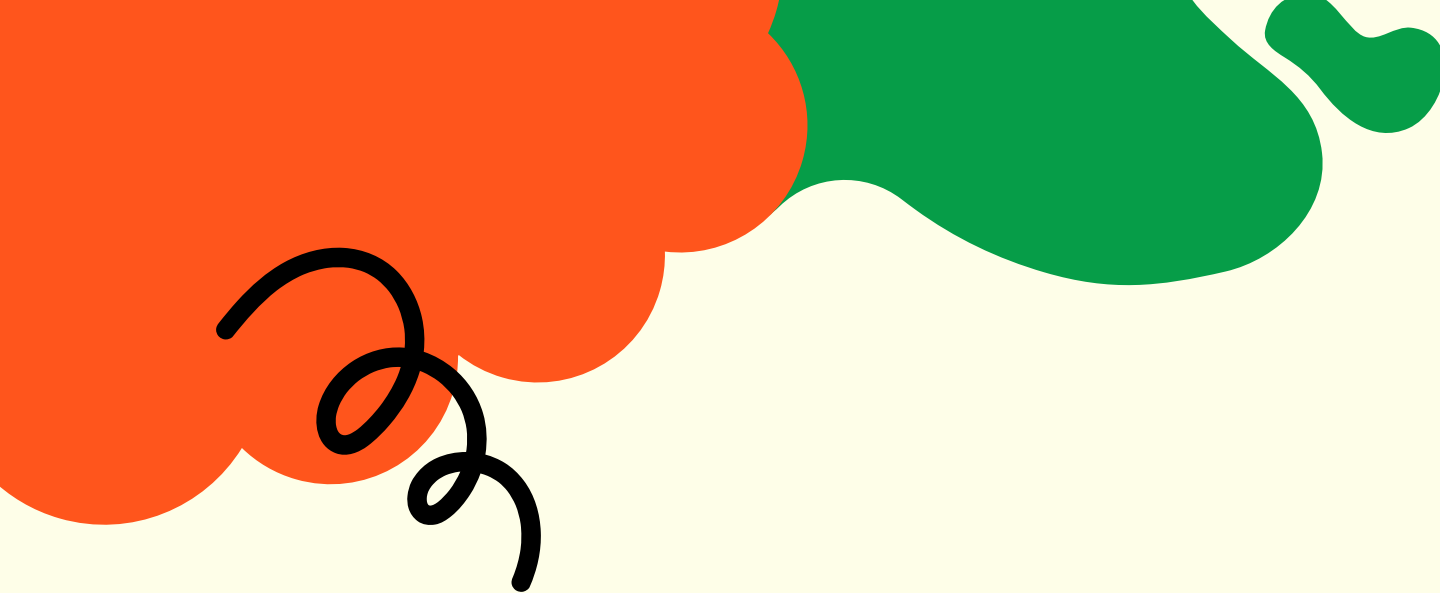
Live Lab: *Moving Forward*



Type this code: `turtle.forward(100)`
Watch the turtle move 100 steps forward,
leaving a bright neon green line behind!

MOVE STEPS





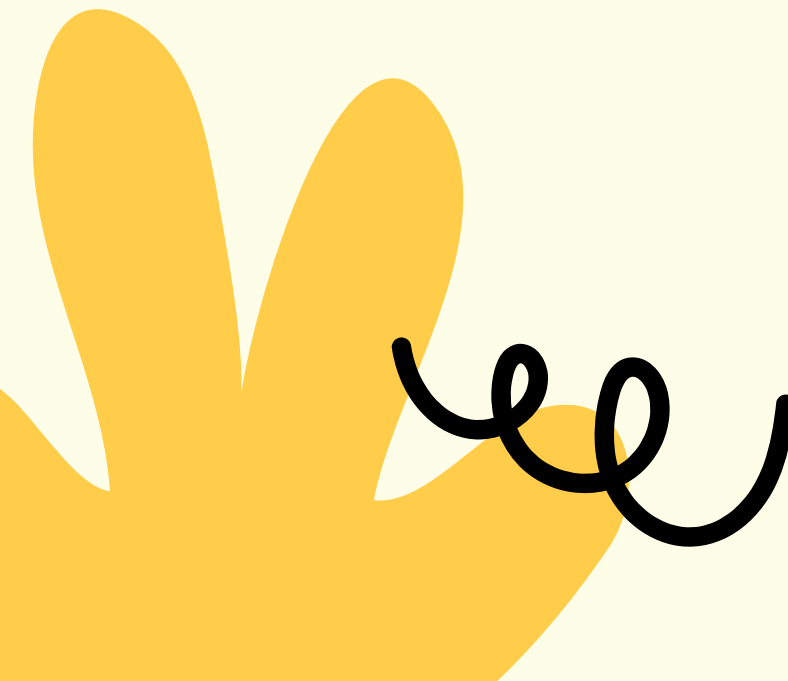
Live Lab: Turning Right

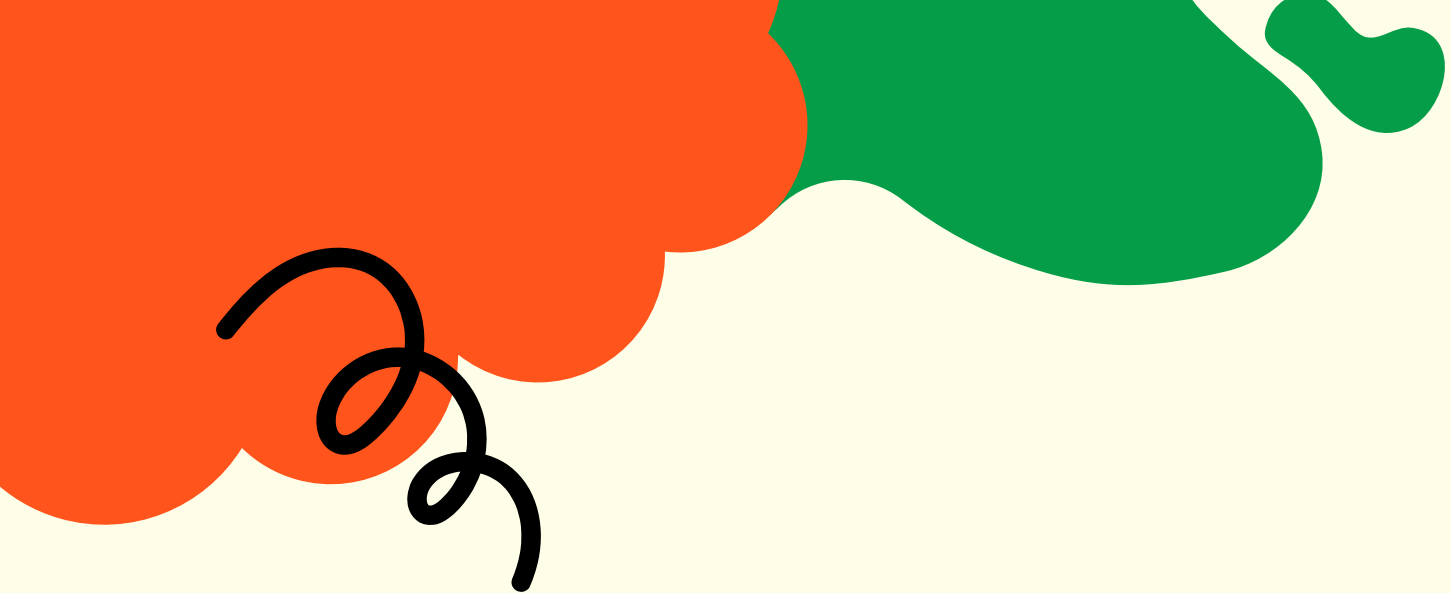
`turtle.right(90)`



Watch the turtle spin right 90 degrees in place!

TURN THE NOSE

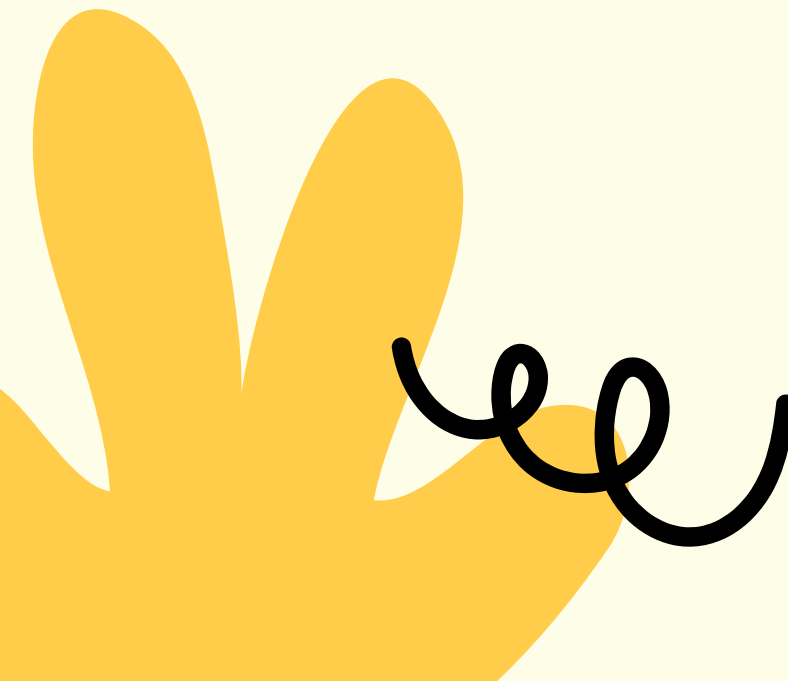




Live Lab: Combining Move and Turn



Watch the magic happen! First, turtle walks forward 100 steps. Then, turtle turns its nose right 90 degrees. Together: `forward(100)`, `right(90)` creates an L-shaped path!





PLANRS Academy

The Square Box

Architect Challenge

Time to build your first shape!



Practice Puzzle 1: Movement Only

What happens when we only use forward commands without any turns? Let's find out what goes wrong!

```
turtle.forward(100) turtle.forward(100)
```

Uh-oh! Turtle ran away off the screen!



Practice Puzzle 2: Syntax Help

Sometimes our code has small mistakes like missing dots or brackets. When this happens, turtle gets confused and shows an error message!

ERROR:

```
turtle.forward(100  
turtle.right(90
```

Hint: Let's fix the brackets so turtle understands!



Practice Puzzle 3: Perfect Square

```
turtle.forward(100)
turtle.right(90)
turtle.forward(100)
turtle.right(90)
turtle.forward(100)
turtle.right(90)
turtle.forward(100)
turtle.right(90)
Great! You built a perfect square!
```





Quick Quiz: Commands Check

Which command spins turtle's nose?



What shape has 4 sides and 90° turns?



Arrange: forward → right → forward



Summary & Home Quest

Code is like dance steps followed one by one

Use forward and right turns to draw shapes

Simple shapes build festive Rangoli art

Home Quest: Find square patterns at home and write step-and-turn instructions





Let's Keep
Creating

Thank You!

